

3. (2.5 points)

Given the following information as of December 31, 2018:

Accident Year	Earned Premium (\$000)	Incremental Paid Loss (\$000) as of (months)		
		12	24	36
2016	800	320	220	80
2017	600	300	200	
2018	400	280		

- Assume there is no loss development beyond 36 months.

a. (2 points)

Calculate the total Neuhaus loss ratio claims reserve estimate.

b. (0.5 point)

Describe why the Neuhaus method may not be appropriate for the data in the table above.

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QUESTION 3

TOTAL POINT VALUE: 2.5

LEARNING OBJECTIVE(S): A1, A2

SAMPLE ANSWERS

Part a: 2.0 points

Sample 1

Neuhaus credibility weights CL and BF estimates using $z = ELR * \pi$

$$m_1 = (320 + 300 + 280) / (800 + 600 + 400) = 0.5$$

$$m_2 = (220 + 200) / (800 + 600) = 0.3$$

$$m_3 = (80) / (800) = 0.1$$

$$ELR = \sum m_i = 0.90$$

$$p_1 = m_1 / ELR = 0.556$$

$$p_2 = (m_1 + m_2) / ELR = 0.889$$

$$p_3 = 1$$

<u>AY</u>	<u>CL Reserve</u>	<u>BF Reserve</u>	<u>z</u>	<u>Neuhaus Reserve</u>
2016	0	0	0.90	0
2017	62.43	59.94	0.80	61.932
2018	223.60	159.84	0.50	191.720
				253.652 (\$000)

Example calcs for AY 2017:

- CL Reserve = $(300 + 200) / 0.889 - (300 + 200)$
- BF Reserve = $600 * (0.90) (1 - 0.889)$
- $Z = 0.90 (0.889)$
- Neuhaus Reserve = $0.80 (62.43) + (1 - 0.8) (59.94)$

Sample 2

	12	24	36	L/R
Mk	.5	.3	.1	.9
Pk	.556	.889	1	
Incremental payout	.556	.333	.111	

AY	ULT ind	ULT coll	Pk * L/R = zNH	ULT NH	Reserve NH
2016	620	620	0	620	0
17	562	560	.8	561.6	61.6
18	504	440	.5	472	192
					253.6

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Sample 3

Neuhaus: $z_k = p_k * ELR$

$$m_{12} = (320 + 300 + 280) / (800 + 600 + 400) = 0.5$$

$$m_{24} = (220 + 200) / (800 + 600) = 0.3$$

$$m_{36} = (80) / (800) = 0.1$$

$$ELR = 0.5 + 0.3 + 0.1 = 0.9 = 90\%$$

$$p_{12} = 0.5 / 0.9 = 55.569\%$$

$$p_{24} = 0.3 / 0.9 = 33.33\%$$

$$p_{36} = 0.1 / 0.9 = 11.11\%$$

$$R_{2017}^{coll} = 11.11\% * 90\% * 600 = 60$$

$$R_{2018}^{coll} = (11.11\% + 33.33\%) * 90\% * 400 = 160$$

$$R_{2017}^{ind} = (500 * 11.11\%) / 88.89\% = 62.5$$

$$R_{2018}^{ind} = (280 * 44.44\%) / 55.56\% = 224$$

$$z_{2017} = 0.8 = 0.9 * 8/9$$

$$z_{2018} = 0.5 = 0.9 * 5/9$$

$$R_{Neuhaus_{2017}} = 0.8 * 62.5 + 0.2 * 60 = 62$$

$$R_{Neuhaus_{2018}} = 0.5 * 224 + 0.5 * 160 = 192$$

$$\text{Total Reserve} = 62 + 192 = 254$$

Part b: 0.5 point

Sample Responses

- It might not be appropriate because it assumes a constant ELR for all AY's, and as the book shrinks the incremental LR's seem to be increasing
Incremental LR @ 12 months
2016 = 0.4
2017 = 0.5
2018 = 0.7
- Neuhaus uses a common LR for all years to compute the credibility factor z but here loss ratio seems to be changing largely year over year as EP declines so using one LR in the credibility weight (as well as to estimate R_{coll}) is not appropriate.
- WN assume constant ELR across AY's.
The data exhibits increasing trend for ELR's
WN is not appropriate

AY	OLEP	p_i	Used Up P	RL	$ELR_i = RL / \text{Used Up P}$
2016	800	1.0	800	620	$=620/800 = 77.5\%$
2017	600	0.889	533	500	93.8%
2018	400	0.556	222	280	126.1%

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- Using an expected loss ratio should mean you expect a consistent loss ratio and/or exposure base. This doesn't seem to be the case here. Premiums have dropped 50% in 2 years, meanwhile the losses have not dropped substantially meaning loss ratios appear to be increasing. This actually has the effect of giving high credibility to a shrinking line.
- Neuhaus works for stable business which is not the case here as the premium is decreasing from 2016-2018
- Premium has been declining every year, may indicate change in mix of business. Neuhaus gives weight to the collective loss ratio reserve which assumes a constant loss ratio every year. Declining premium may indicate that constant loss ratio cannot be assumed.
- The Neuhaus may not be appropriate because there is a decline in earned prem over the 3 accident years, and the credibility calculation may distort due to the weight placed on the collected method.
- The premium volume is shrinking over time. This may indicate historical premium/losses are not representative of current/future premium and loss. The Neuhaus method assumes the overall ELR, calculated from the data, is appropriate for all years. This may not be the case here if the mix of business / overall loss ratio is changing over time.

EXAMINER'S REPORT

Candidates were expected to be able to calculate Neuhaus reserve estimates, understand the mechanics of the method, and test the suitability of this model to the given data set.

Part a

The candidate was expected to know how to estimate reserves according to the Neuhaus method.

Common mistakes included:

- Not using cumulative paid losses in the 2017 year when estimating individual reserves.
- Only calculating reserves for a single year (2017 or 2018)
- Calculating a non-zero reserve for the 2016 accident year
- When estimating reserves for either the collective or individual method, neglecting to multiply by q (the loss ratio reserve factor)
- Deriving payout factors and/or individual reserve estimates based on chain ladder LDF method
- Calculating reserves based on the expected loss ratio method rather than the BF reserve as the collective reserve estimate
- Misinterpreting the claims triangle as cumulative instead of incremental losses

Part b

Candidates were expected to identify patterns in the premium or loss and their fit to the Neuhaus model assumptions. A variety of responses were accepted in regard to tying the observation back to the Neuhaus method, including responses where candidates commented on either how the Neuhaus method works or underlying assumptions of the method.

Common mistakes included:

- Not specifically commenting on the trends evident in the data.
- Not tying the trends in the data back to the Neuhaus method.